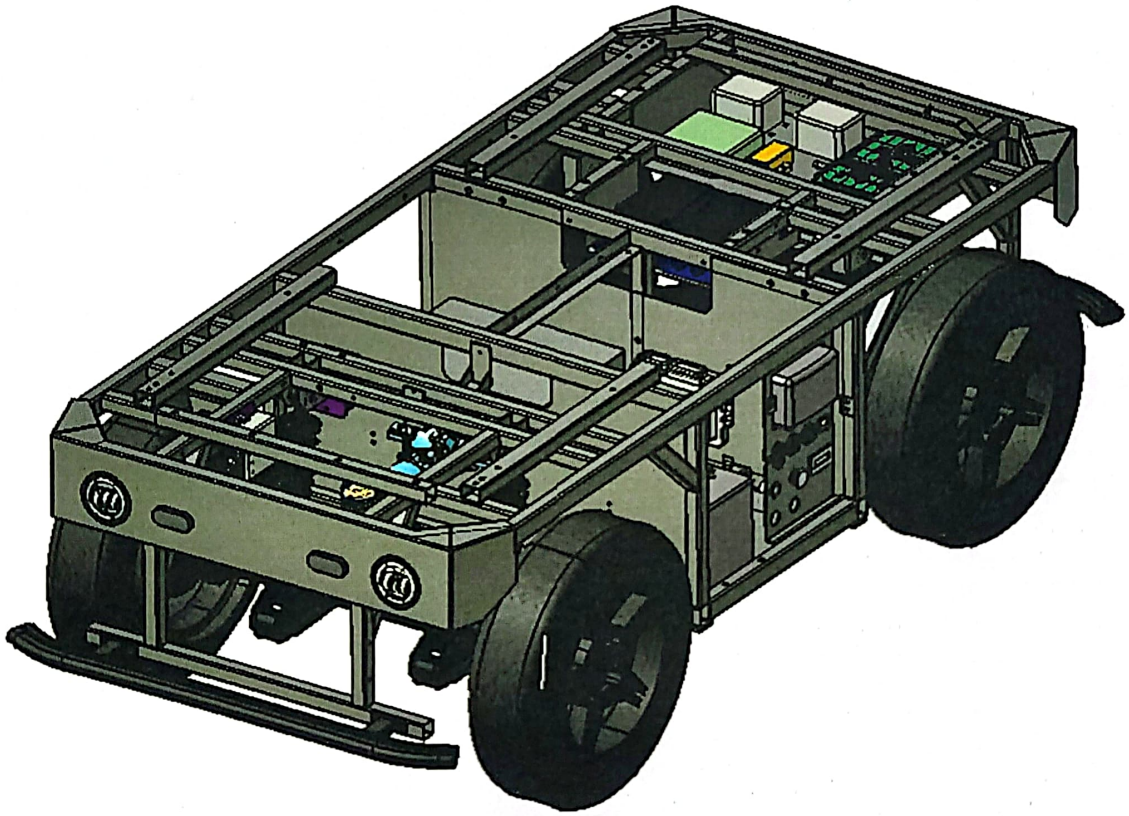
WITH:US-S1

WITH:US-Small-size utility robot 1 User Manual

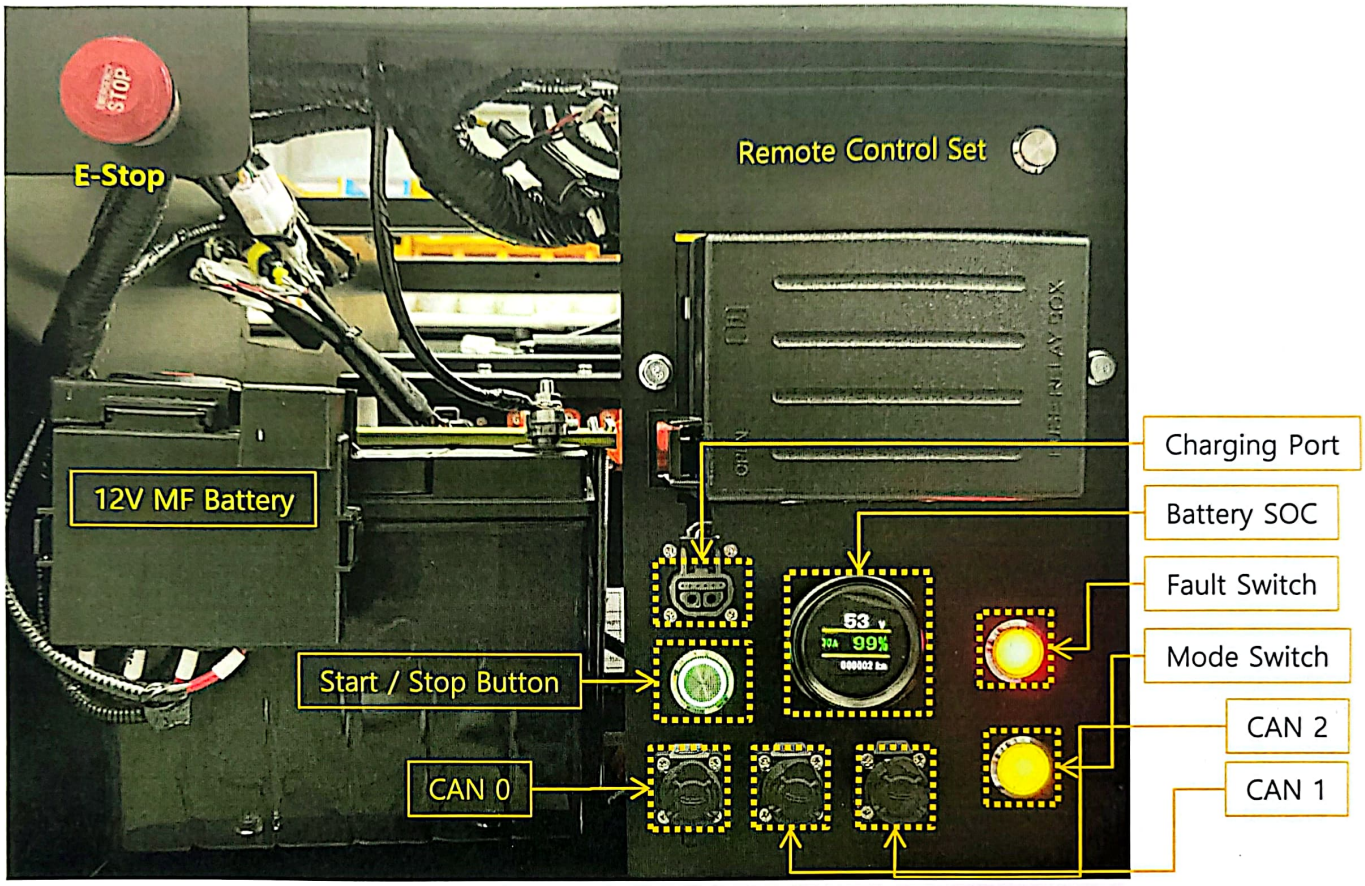


Version 1.0

Unmanned Solution

Vehicle Unit Position

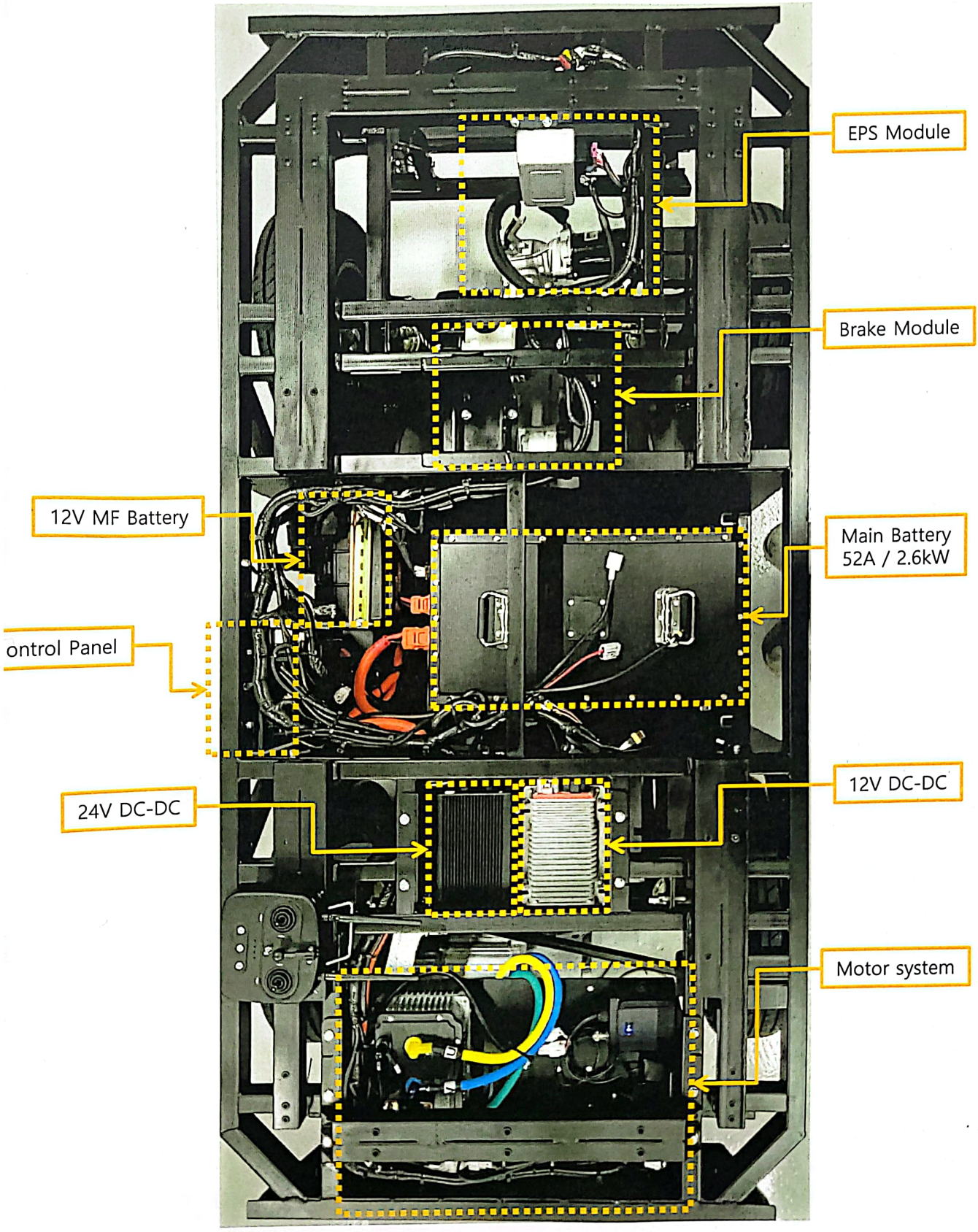
Indicator Lamp & Battery State Of Charge



Name	Disciption	Remark
Start / Stop Switch	• 차량 시동 On / Off Switch (On상태에서 점등, 5초 후에 Off)	
Battery SOC	• Battery Full Charge : 54.0V • Battery Low Level : 47.0(자동 전원 차단) ※ 48.0V 이하로 내려가지 않게 충전을 해주시기 바랍니다.	
Fault Switch	• 차량이 고장이나 연결 문제로 주행 불가능(Red) • 차량 주행 가능(Green)	
Mode Switch	• Manual Mode(Yellow) • Autonomous mode(Green)	
CAN0	• 하단 CAN Protocol Page 참조	하단 CAN Protocol Page참조
CAN1	• 하단 CAN Protocol Page 참조	
CAN2	• Brake Module Support CAN Line	Only CAN ID 142, 143
Emergency Switch	• 차량 비상정지 스위치로 시동 후 출발 전에 해제 바랍니다. • 차량 주행 완료 후 비상정지 상태로 변경 바랍니다.	
Remote control set	• Remote control 재연결때 사용.	
12V MF Battery	• Main Battery에서 자동 충전 • 12V System 보조 Battery	

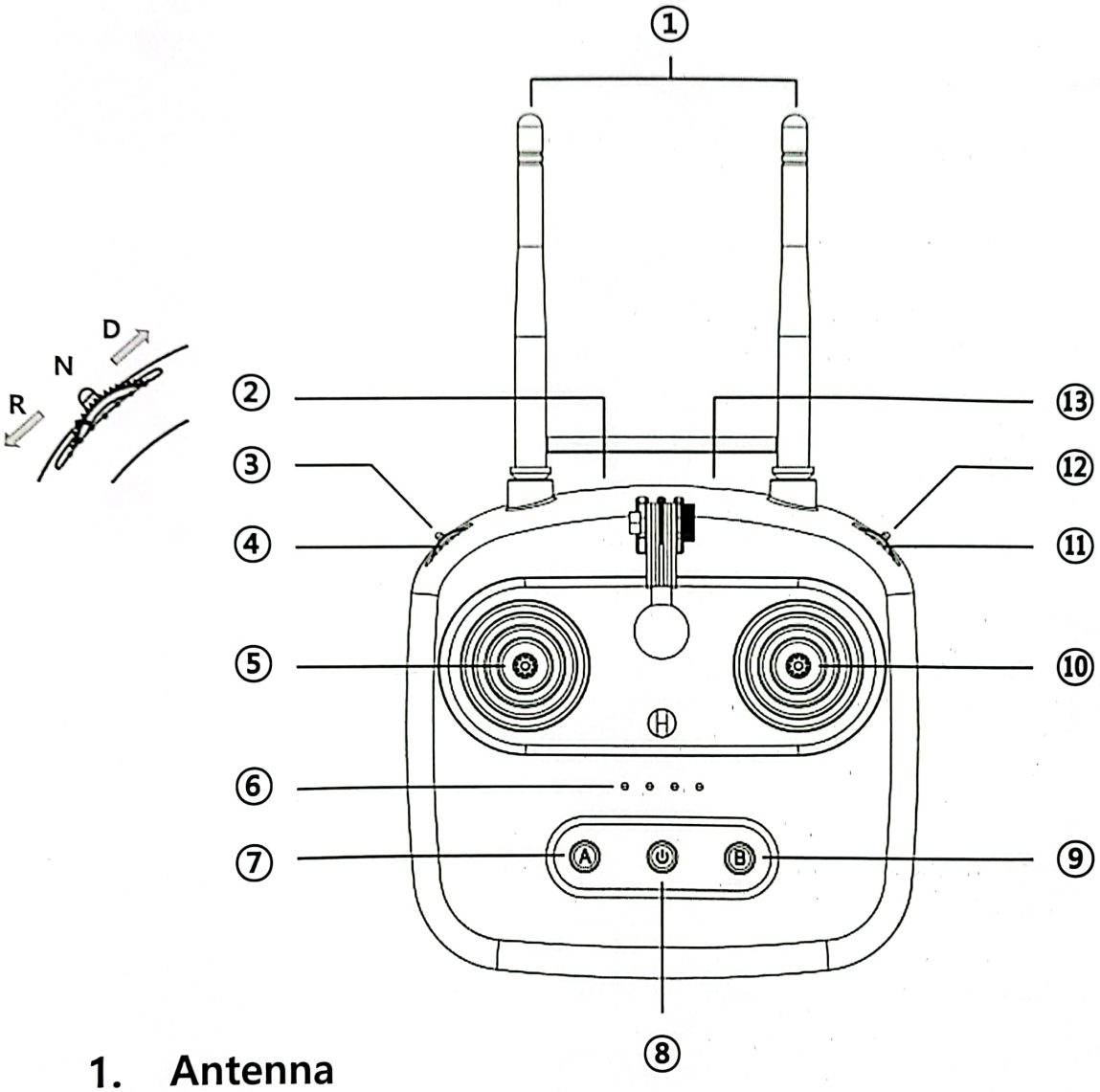
Vehicle Unit Position

차량 부품 배치도



Unmanned Solution

Vehicle Unit Position



1. Antenna
2. Non-function
3. Toggle the third gear switch
4. Dial wheel - Drive torque ring switch
5. Left Rocker – ↑: Throttle, ↓: Brake lever
6. Remote control battery indicator
7. Button A – Autopilot command switch
8. Power Switch
9. Button B – Motor brake switch
10. Right rocker - ←: Left steering, →: Right steering
11. Dial wheel – Head light switch
12. Toggle Horn switch
13. Charging port

CAN ID List

CAN 0 ID

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
BMS_1A3h		normal			0x1a3			cycle			100			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_Pwr30s	Instantaneous maximum discharge power of the power battery	Intel	6	48	cycle	16	unsigned	0.1	0.0	0.0	6553.5	0x00	0xffff	kw	
BMS_Pwr	Current maximum discharge power of power battery	Intel	4	32	cycle	16	unsigned	0.1	0.0	0.0	6553.5	0x00	0xffff	kw	
BMS_ChgPwr30s	Instantaneous charging power of the power battery	Intel	2	16	cycle	16	unsigned	0.1	0.0	0.0	6553.5	0x00	0xffff	kw	
BMS_ChgPwr	Current maximum charging power of power battery	Intel	0	0	cycle	16	unsigned	0.1	0.0	0.0	6553.5	0x00	0xffff	kw	

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
BMS_2B1h		normal			0x2b1			cycle			100			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_HVBatLowestTemCellNum	- High Voltage Battery Lowest Temperature Cell Number Valid value range 1 ~ 250 - 0xFE : Abnormal - 0xFF : Invalid	Intel	4	32	cycle	8	unsigned	1.0	0.0	0.0	250.0	0x00	0xff		
BMS_HVBatHighestTemCellNum	- High Voltage Battery highest Temperature Cell Number Valid value range 1 ~ 250 - 0xFE : Abnormal - 0xFF : Invalid	Intel	5	40	cycle	8	unsigned	1.0	0.0	0.0	250.0	0x00	0xff		
BMS_HVBatHighestTem	High Voltage Battery highest Temperature	Intel	1	8	cycle	8	unsigned	1.0	-40.0	-40.0	125.0	0x00	0xff	℃	
BMS_HVBatLowestTem	High Voltage Battery Lowest Temperature	Intel	0	0	cycle	8	unsigned	1.0	-40.0	-40.0	125.0	0x00	0xff	℃	

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
VCU_5A3h		normal			0x5a3			cycle			20			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
VCU_Ctrl_AcDc	0x00 : HV OFF 0x01 : HV ON 0xFF : Invalid air conditioning relay control	Intel	1	8	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		
VCU_PowerOn_BMS	0 : Positive pull-in 1 : Main positive disconnect	Intel	0	0	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		

CAN ID List

CAN 0 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
BMS_2B2h	normal	0x2b2	cycle	1000	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_HVBatCell Tem_No8	High Voltage Battery Cell Temperature_No.8	Intel	7	56	cycle	8	unsigned	1.0	-40.0	-40.0	210.0	0x00	0xff		
BMS_HVBatCell Tem_No7	High Voltage Battery Cell Temperature_No.7	Intel	6	48	cycle	8	unsigned	1.0	-40.0	-40.0	210.0	0x00	0xff		
BMS_HVBatCell Tem_No6	High Voltage Battery Cell Temperature_No.6	Intel	5	40	cycle	8	unsigned	1.0	-40.0	-40.0	210.0	0x00	0xff		
BMS_HVBatCell Tem_No5	High Voltage Battery Cell Temperature_No.5	Intel	4	32	cycle	8	unsigned	1.0	-40.0	-40.0	210.0	0x00	0xff		
BMS_HVBatCellTe m_No4	High Voltage Battery Cell Temperature_No.4	Intel	3	24	cycle	8	unsigned	1.0	-40.0	-40.0	210.0	0x00	0xff		
BMS_HVBatCellTe m_No3	High Voltage Battery Cell Temperature_No.3	Intel	2	16	cycle	8	unsigned	1.0	-40.0	-40.0	210.0	0x00	0xff		
BMS_HVBatCellTe m_No2	High Voltage Battery Cell Temperature_No.2	Intel	1	8	cycle	8	unsigned	1.0	-40.0	-40.0	210.0	0x00	0xff		
BMS_HVBatCellTe m_No1	High Voltage Battery Cell Temperature_No.1	Intel	0	0	cycle	8	unsigned	1.0	-40.0	-40.0	210.0	0x00	0xff		

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
BMS_2A5h	normal	0x2a5	cycle	1000	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_HVBatCellVol No16	High Voltage Battery Cell Voltage_No.16	Intel	6	48	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol _No15	High Voltage Battery Cell Voltage_No.15	Intel	4	32	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol No14	High Voltage Battery Cell Voltage_No.14	Intel	2	16	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol No13	High Voltage Battery Cell Voltage_No.13	Intel	0	0	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
BMS_2A4h	normal	0x2a4	cycle	1000	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_HVBatCellVol No12	High Voltage Battery Cell Voltage_No.12	Intel	6	48	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol No11	High Voltage Battery Cell Voltage_No.11	Intel	4	32	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol No10	High Voltage Battery Cell Voltage_No.10	Intel	2	16	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol No9	High Voltage Battery Cell Voltage_No.9	Intel	0	0	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
BMS_2A3h	normal	0x2a3	cycle	1000	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_HVBatCellVol No8	High Voltage Battery Cell Voltage_No.8	Intel	6	48	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol _No7	High Voltage Battery Cell Voltage_No.7	Intel	4	32	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol No6	High Voltage Battery Cell Voltage_No.6	Intel	2	16	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol No5	High Voltage Battery Cell Voltage_No.5	Intel	0	0	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	

CAN ID List

CAN 0 ID

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
BMS_2A2h		normal			0x2a2			cycle			1000			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_HVBatCellVol_No4	High Voltage Battery Cell Voltage_No.4	Intel	6	48	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol_No3	High Voltage Battery Cell Voltage_No.3	Intel	4	32	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol_No2	High Voltage Battery Cell Voltage_No.2	Intel	2	16	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBatCellVol_No1	High Voltage Battery Cell Voltage_No.1	Intel	0	0	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
BMS_2A1h		normal			0x2a1			cycle			100			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_HVBat LowestVolCellNum	0x0 - 0xFF : Cell Number	Intel	4	32	cycle	8	unsigned	1.0	0.0	0.0	250.0	0x00	0xff		
BMS_HVBat HighestVolCellNum	0x0 - 0xFF : Cell Number	Intel	5	40	cycle	8	unsigned	1.0	0.0	0.0	250.0	0x00	0xff		
BMS_HVBat HighestCellVol	High Voltage Battery highest Cell Voltage	Intel	2	16	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	
BMS_HVBat LowestCellVol	High Voltage Battery Lowest Cell Voltage	Intel	0	0	cycle	16	unsigned	1.0	0.0	0.0	5000.0	0x00	0xffff	mV	

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
BMS_A0h		normal			0xa0			cycle			100			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
BMS_HV DisplaySOH	Show SOH	Intel	7	56	cycle	8	signed	1.0	0.0	0.0	100.0	0x00	0xff	%	
BMS_Sys_Flt	error code	Intel	6	48	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		
BMS_Charge _StsCc	Slow charging connection status 0x00 : Connect 0x01 : Disconnect	Intel	5	44	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
BMS_Charge _StsCc2	Fast charging connection status 0x00 : Connect 0x01 : Disconnect	Intel	5	43	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
BMS_Sys_Sts	0x0 : Normal 0x1 : Level 1 fault(Alarm) 0x2 : Level 2 Fault(Power limit) 0x3 : Level 3 Fault (power limit is 0, active shutdown)	Intel	5	40	cycle	3	unsigned	1.0	0.0	0.0	7.0	0x00	0x7		
BMS_HVBatSOC	High Voltage Battery State Of Charge High voltage battery SOC state	Intel	4	32	cycle	8	unsigned	0.4	0.0	0.0	100.0	0x00	0xff	%	
BMS_HVBatVol	High Voltage Battery Voltage High voltage battery current voltage	Intel	2	16	cycle	16	unsigned	0.1	0.0	0.0	1000.0	0x00	0xffff	V	
BMS_HVBatCrnt	High Voltage Battery Current High voltage battery current	Intel	0	0	cycle	16	unsigned	0.1	-1000.0	-1000.0	1000.0	0x00	0xffff	A	

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CAN ID List

CAN 0 ID

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
BMS_1A2h		normal			0x1a2			cycle			100			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Charge_Relay_Sts	0x00 : Open 0x01 : Close charging relay status	Intel	6	51	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
AcDc_Relay_Sts	0x00 : Open 0x01 : Close Air conditioning relay status	Intel	6	52	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
BMS_WeakUpSig	BMS wake-up source 0x01 : keyon 0x02 : A+ 0x04 : CP	Intel	0	0	cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
BMS_KeyOn	Normal voltage	Intel	2	16	cycle	8	unsigned	0.1	0.0	0.0	25.5	0x00	0xff	V	
BMS_Req _HvDown	BMS requests high voltage power off 0x00 : No request 0x01 : Requests high voltage power off	Intel	5	40	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Bms_SelfChk_Sts	BMS self-test status 0x00 : Self-test in progress 0x01 : Self-test successful 0x02 : Self-test failed	Intel	5	43	cycle	3	unsigned	1.0	0.0	0.0	7.0	0x00	0x7		
Heat_Relay_Sts	Heating relay status 0x00 : Open 0x01 : Close	Intel	5	46	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Precharge_Sts	Precharge status 0x00 : Invalid 0x01 : Precharge in progress 0x02 : Precharge successful 0x03 : Precharge failed	Intel	6	49	cycle	2	unsigned	1.0	0.0	0.0	3.0	0x00	0x3		
Charge_Sts	0x00 : Non-charging state 0x01 : Charging 0x02 : Heating 0x03 : Heating while charging 0x04 : Charging completed (when charging is completed, a cha rging failure is detected and chargi ng is completed) 0x05: Charging Fault	Intel	6	53	cycle	3	unsigned	1.0	0.0	0.0	7.0	0x00	0x7		
BMS_Heat_Beat	Heartbeat signal	Intel	7	56	cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Precharge_ Relay_Sts	Precharge relay status 0x00 : Open 0x01 : Close	Intel	6	48	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Parent_Relay_Sts	Main positive relay status 0x00 : Open 0x01 : Close	Intel	5	47	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		

CAN ID List

CAN 1 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
VCU_Vehicle_Status_1	normal	0x303	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Vehicle_Status _MsgCntr	Heartbeat Cycle 0-15	Intel	7	60	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Drive_Mode_State	0 : Remote control mode 1 : Represents the AD mode 2 : Indicates parallel driving mode 3 : Indicates semi-autonomous driving mode (AD or parallel driving with remote control. such as ad(parallel driving) steering control, Remote control drive brake)	Intel	1	8	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		1 : 501 – 506 2 : 511 – 516
Vehicle_Soc	Vehicle residual electricity %	Intel	6	48	Cycle	8	unsigned	1.0	0.0	0.0	100.0	0x00	0xff	%	
VCU_Speed_Req	VCU Speed Request km/h	Intel	2	16	Cycle	16	unsigned	0.1	-80.0	-80.0	80.0	0x00	0xffff	Km/h	
Clamping _Brake_Status	Brake holding state 0 : loose 1 : Tight	Intel	0	3	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Vehicle_Gear	Vehicle gearing status 0 : P gear 1 : D gear 2 : N gear 3 : R gear	Intel	0	0	Cycle	2	unsigned	1.0	0.0	0.0	3.0	0x00	0x3	N/A	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
Parallel_Control_Flag	normal	0x511	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Parallel _Flag_MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Parallel_Control _Request_Flag	0 : Invalid 1 : Valid	Intel	0	0	Cycle	4	unsigned	1.0	0.0	0.0	1.0	0x00	0xf	N/A	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
Parallel_Control_Body	Normal	0x516	cycle	50	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Parallel_ BrakeLight_Switch	0 : Light off 1 : Light on	Intel	1	8	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off
Parallel _Body_MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf	N/A	
Parallel _Hom_Switch	0 : Horn off 1 : Horn on	Intel	0	2	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off
Parallel _Headlight_Switch	0 : Light off 1 : Light on	Intel	0	3	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off
Parallel_Right _Light_Switch	0 : Right turn Light off 1 : Right turn Light on	Intel	0	1	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off
Parallel_Left _Light_Switch	0 : Left turn Light off 1 : Left turn Light on	Intel	0	0	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off

CAN ID List

CAN 1 ID

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
Parallel_Control_Steering		normal			0x512			cycle			20			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Parallel_ Steering_MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Parallel_ Steering_Valid	0 : Invalid 1 : Controllable steering	Intel	0	0	Cycle	4	unsigned	1.0	0.0	0.0	1.0	0x00	0xf	N/A	0x1: Valid 0x0: Invalid
Parallel_Steering _Angle_Cmd	From -30 to +30 degrees Corner	Intel	4	32	Cycle	16	unsigned	0.1	-30.0	-30.0	30.0	0x00	0xffff	deg	

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
Parallel_Control_Accelerate		normal			0x514			cycle			20			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Parallel_Accelerate _MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Parallel_Acc_De	Parallel Drive -5 ~ 5m/ss Acceleration Request	Intel	4	32	Cycle	8	unsigned	0.1	-5.0	-5.0	5.0	0x00	0xff	m/ss	
Parallel_driving _Speed_Control	Vehicle requested speed km/h	Intel	6	48	Cycle	16	unsigned	0.1	0.0	0.0	20.0	0x00	0xffff	km/h	
Parallel_driving _Torque_Control	Drive throttle opening 0~100%	Intel	5	40	Cycle	8	unsigned	1.0	0.0	0.0	100.0	0x00	0xff	%	
Parallel _Accelerate_Gear	0 : P gear 1 : D gear 2 : N gear 3 : R gear	Intel	3	24	Cycle	8	unsigned	1.0	0.0	0.0	3.0	0x00	0xff		
Parallel_Accelerate _Work_Mode	0 : Torque control 1 : Speed control 2 : Accel and deceleration control	Intel	2	16	Cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff		
Parallel _Accelerate_Valid	0 : Invalid 1 : Controllable drive	Intel	0	0	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
Parallel_Control_Brake		normal			0x513			cycle			20			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Parallel _DBS_MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Parallel_ BrakePressure_Cmd	Brake pedal (0~100%) 0~8Mpa pressure request	Intel	1	8	Cycle	8	unsigned	1.0	0.0	0.0	100.0	0x00	0xff	%	
Parallel_DBS_Valid	0 : Invalid 1 : Brake controllable	Intel	0	0	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		0x1: Valid 0x0: Invalid

CAN ID List

CAN 1 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
VCU_EPS_Control_Request	normal	0x314	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
VCU_EPS _StrAngle_Req	Directional end angle request	Motorola MSB	1	15	Cycle	16	unsigned	1.0	0.0	-720.0	720.0	0x00	0xffff	deg	
VCU_EPS _CtrlEnable	EPS control enabled 0 : Stops 1 : Works	Motorola MSB	0	0	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	
VCU_EPS _Calibration_Req	Calibration control bit 0 : Invalid 1 : Set position to the center	Motorola MSB	0	2	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
AD_Control_Body	Normal	0x506	cycle	50	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
AD_Brake_Light	0 : Brake light off 1 : Brake light on	Intel	1	8	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off
AD_Body_MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
AD_Horn_Control	0 : Horn off 1 : Horn on	Intel	0	2	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off
AD_HeadLight	0 : Headlights off 1 : Headlight on	Intel	0	3	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off
AD_Right _Turn_Light	0 : Right turn light off 1 : Right turn light on	Intel	0	1	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off
AD_Left _Turn_Light	0 : Left turn light off 1 : Left turn light on	Intel	0	0	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	0x1: Turn On 0x0: Turn Off

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
VCU_Vehicle_Status_2	normal	0x304	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Vehicle_ Status_2_MsgCntr	Heartbeat Cycle 0-15	Intel	7	60	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Vehicle_ Steering_Angle	Vehicle wheel end angle	Intel	4	32	Cycle	10	unsigned	0.1	-35.0	-35.0	67.3	0x00	0x3ff	deg	
Vehicle_ Brake_Pressure	Vehicle brake line pressure	Intel	2	16	Cycle	16	unsigned	0.01	0.0	0.0	8.0	0x00	0xffff	Mpa	
Vehicle_Speed	Vehicle speed km/h	Intel	0	0	Cycle	16	unsigned	0.1	-80.0	-80.0	80.0	0x00	0xffff	km/h	

CAN ID List

CAN 1 ID

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
AD_Control_Accelerate		normal			0x504			cycle			20			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
AD_Accelerate _MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
AD_Acc_De	AD -5~5m/ss Accel Request	Intel	4	32	Cycle	8	unsigned	0.1	-5.0	-5.0	5.0	0x00	0xff	m/ss	
AD_Speed _Control	Vehicle requested speed (km/h)	Intel	6	48	Cycle	16	unsigned	0.1	0.0	0.0	80.0	0x00	0xffff	km/h	
AD_Torque _Control	Drive throttle opening 0~100% Corresponding torque 0~20N.m	Intel	5	40	Cycle	8	unsigned	1.0	0.0	0.0	100.0	0x00	0xff	%	
AD_ Accelerate_Gear	0 : P gear 1 : D gear 2 : N gear 3 : R gear	Intel	3	24	Cycle	8	unsigned	1.0	0.0	0.0	3.0	0x00	0xff	N/A	0x3: R 0x2: N 0x1: D 0x0: P
AD_Accelerate _Work_Mode	0 : Torque control mode 1 : Speed control	Intel	2	16	Cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff	N/A	0x3: Acc_Control 0x1: Speed_Control 0x0: Torque_Control
AD_ Accelerate_Valid	0 : Invalid 1 : Controllable drive	Intel	0	0	Cycle	4	unsigned	1.0	0.0	0.0	1.0	0x00	0xf	N/A	0x1: Valid 0x0: Invalid

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
AD_Control_Brake		normal			0x503			cycle			20			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
AD_DBS_MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
AD_BrakePressure _Cmd	Brake pedal (0~100%) 0-8Mpa pressure request	Intel	1	8	Cycle	8	unsigned	1.0	0.0	0.0	100.0	0x00	0xff	%	
AD_DBS_Valid	0 : Invalid 1 : Controllable brake	Intel	0	0	Cycle	4	unsigned	1.0	0.0	0.0	1.0	0x00	0xf	N/A	0x1: Valid 0x0: Invalid

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
AD_Control_Steering		normal			0x502			cycle			20			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
AD_Steering _MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
AD_Steering _Angle_Cmd	AD mode -30° to +30° corner request	Intel	4	32	Cycle	16	unsigned	0.1	-30.0	-30.0	30.0	0x00	0xffff	deg	
AD_ Steering_Valid	0 : Invalid 1 : Controllable steering	Intel	0	0	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf	N/A	0x1: Valid 0x0: Invalid

CAN ID List

CAN 1 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
AD_Control_Flag	normal	0x501	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
AD_Flag_MsgCntr	Heartbeat Cycle 0-15	Intel	0	4	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
AD_Control_Request_Flag	0 : Invalid. 1 : Valid.	Intel	0	0	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf	N/A	0x1: Valid 0x0: Invalid

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
VCU_Vehicle_Diagnosis	normal	0x301	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Vehicle_Voltage	Battery voltage	Intel	2	22	Cycle	10	unsigned	0.1	0.0	0.0	90.0	0x00	0x3ff	V	
Light_states_Brake	0 : Brake light off 1 : Brake light on	Intel	5	40	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
EPS_State	0 : Not fault 1 : Fault	Intel	1	10	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	
Oil_pot_State	0 : Oil level normal 1 : Abnormal	Intel	5	41	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	
Vehicle_Diagnosis_MsgCntr	Heartbeat Cycle 0-15	Intel	7	60	Cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Horn_State	0 : Horn off 1 : Horn on	Intel	5	43	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
HeadLight_State	0 : Headlight off 1 : Headlight on	Intel	1	15	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Right_Turn_Light_State	0 : Right turn light off 1 : Right turn light on	Intel	6	48	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Left_Turn_Light_State	0 : Left turn light off 1 : Left turn light on	Intel	4	32	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
R_Touch_Switch_State	0 : Sensor not trigger 1 : Sensor trigger	Intel	1	13	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
F_Touch_Switch_State	0 : Sensor not trigger 1 : Sensor trigger	Intel	1	12	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
BMS_State	0 : Normal 1 : Fault.	Intel	1	9	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Emergency_Button_State	0 : Not pressed 1 : Pressed	Intel	0	0	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Parallel_State	0 : Parallel driving 1 : Not parallel driving	Intel	1	8	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	
DBS_State	0 : Not fault 1 : Fault	Intel	0	7	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	
AD_State	0 : AD signal received 1 : AD signal not received	Intel	0	6	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	
Remote_State	0 : Not fault 1 : Fault	Intel	0	5	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	
Motor_State	0 : Not fault 1 : Fault	Intel	0	1	Cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	

CAN ID List

CAN 1 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
RemoteT10_Control_Shake_2	normal	0x10b	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Remote_Y1_Longitudinal_Control	0~450 corresponds to 0~100% throttle opening request, indicating acceleration; 0~-450 corresponds to 0-100% braking force request, indicating deceleration.	Intel	4	32	cycle	16	signed	1.0	0.0	-450.0	450.0	0x00	0xffff	N/A	
Remote_X2_Lateral_Control	0~450 corresponds to 0-100% to the right steering angle request, indicating a right turn; 0~-450 corresponds to 0-100% left turn request, and remote indicating a left turn.	Intel	0	0	cycle	16	signed	1.0	0.0	-450.0	450.0	0x00	0xffff	N/A	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
RemoteT10_Control_IO	normal	0x10a	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Remote_F_Horn	control horn	Intel	1	8	cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff		
Remote_D_HeadLight	0 : Headlights off 1, 2 : Headlights on	Intel	5	40	cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff	N/A	
Remote_B_Motor_Holding_Brake	0 : Motor brake released 2 : Motor holding brake	Intel	3	24	cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff	N/A	
Remote_A_Mode_Switch	Autonomous driving mode	Intel	2	16	cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff	N/A	
Remote_C_Speed_Torque	0 : Torque control 1, 2 : Speed control	Intel	4	32	cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff	N/A	
Remote_E_Gear	0 : Reverse gear 1 : Neutral gear 2 : Forward gear	Intel	0	0	cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff	N/A	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
EPS_Status	normal	0x18f	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
EPS_StrAngle_Act	EPS Current angle value	Motorola MSB	1	15	cycle	16	signed	1.0	0.0	-720.0	720.0	0x00	0xffff		
EPS_Temperature	EPS ECU Temperature	Motorola MSB	6	55	cycle	8	signed	1.0	0.0	-128.0	127.0	0x00	0xff	℃	
EPS_Motor_Current	Motor current (mA)	Motorola MSB	3	31	cycle	16	signed	1.0	0.0	-32768.0	32767.0	0x00	0xffff	mA	
EPS_Fault	0 : ECU Not fault 1 : ECU Fault	Motorola MSB	0	1	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
EPS_Calibration_Status	1 EPS median is not calibrated 0 EPS median is calibrated	Motorola MSB	0	2	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
EPS_Work_Mode	0 : Stop mode 1 : Operating mode	Motorola MSB	0	0	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	

CAN 1 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
VCU_DBS_Request	normal	0x154	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
VCU_ABS_Active	ABS work flag 0 : Not working 1 : Working.	Intel	3	24	cycle	8	unsigned	1.0	0.0	0.0	1.0	0x00	0xff		
VCU_DBS_Request_Flag	VCU Brake request Enable bit 0 : Power assistance mode 1 : Wire control mode	Intel	0	0	cycle	8	unsigned	1.0	0.0	0.0	1.0	0x00	0xff		
VCU_DBS_Pressure_Request	Send target pressure	Intel	2	16	cycle	8	unsigned	0.1	0.0	0.0	8.0	0x00	0xff	Mpa	
VCU_DBS_Work_Mode	Exhaust mode 0 : Uline control mode 1 : Exhaust mode	Intel	1	8	cycle	8	unsigned	1.0	0.0	0.0	0.0	0x00	0xff		

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
DBS_Status2	normal	0x143	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
DBS_CheckSum	Checksum = byte0 XOR byte1 XOR byte2 XOR--XOR byte6	Intel	7	56	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		
DBS_RollingCounter	Counts in a loop from 0 to 15, if communication is interrupted, Then start counting again from 0	Intel	6	48	cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
DBS_WaringCode	Warning Type Bit 1 : Emergency stop button closing warning Bit 2 : The parking time is too long, more than 1 minute Bit 3 : CAN1 interrupt warning Bit 4 : CAN2 interrupt warning Bit 5 : Ignition wire disconnection warning	Intel	3	24	cycle	24	unsigned	1.0	0.0	0.0	16777215.0	0x00	0xffffff		
DBS_FaultCode	Fault Type Bit 1 : Overload protection Bit 2 : Over-temperature protection Bit 3 : Overcurrent protection Bit 4 : Power undervoltage fault Bit 5 : Power supply overvoltage fault Bit 6 : Insufficient pressure Bit 7 : Motor failure Bit 8 : Communication failure Bit 9 : Current sampling fault Bit 10 : Driver failure Bit 11 : Magnetic encoding failure Bit 12 : Pressure sensor failure Bit 13 : Pedal position sensor failure Bit 14 : Other faults	Intel	0	0	cycle	24	unsigned	1.0	0.0	0.0	16777215.0	0x00	0xffffff		

CAN 1 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
DBS_Status	normal	0x142	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
DBS_EstopFlag	Emergency stop sign	Intel	6	54	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
DBS_PedaiFlag	Pedal insertion flag 0 : Pedal not inserted 1 : Pedal inserted	Intel	6	55	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
DBS_Ref_Iq	Actual current	Intel	5	40	cycle	8	unsigned	0.5	-20.0	-20.0	107.5	0x00	0xff		
DBS_Work_Mode	Operation mode 0x0 : CAN control 0x1 : Exhaust Mode 0x2 : Pedal Mode	Intel	1	8	cycle	8	unsigned	1.0	0.0	0.0	2.0	0x00	0xff		
BrakePressure ReqACK	Brake pressure request response, DBS execution completion response is valid 0x0 : Not ACK 0x1 : ACK	Intel	2	16	cycle	8	unsigned	1.0	0.0	0.0	1.0	0x00	0xff		
DBS_ RollingCounter	Counts in a loop from 0 to 15, if communication is interrupted, Then start counting again from 0	Intel	6	48	cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
DBS_Park_Warning	Parking time warning signal 0 : Working normally 1 : Parking time more than 1 minute, issue a warning, recommended EPB parking request	Intel	0	6	cycle	2	unsigned	1.0	0.0	0.0	1.0	0x00	0x3		
DBS_ PeadalOpening	Pedal opening and closing (0-100%)	Intel	4	32	cycle	8	unsigned	1.0	0.0	0.0	100.0	0x00	0xff	%	
DBS_CheckSum	Checksum = byte0 XOR byte1 XOR byte2 XOR--XOR byte6	Intel	7	56	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		
DBS_HP_pressure	Front wheel brake system pressure feedback	Intel	3	24	cycle	8	unsigned	0.1	0.0	0.0	8.0	0x00	0xff	Mpa	
DBS_System_Status	System fault status 0 : System No Fault 1 : System Warning 2 : System Fault	Intel	0	0	cycle	2	unsigned	1.0	0.0	0.0	3.0	0x00	0x3	N/A	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
MCU_Drive_Motor_Feedback_Msg	normal	0x60	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Clamping_ brake_status	Brake status 0 : Release 1 : Hold brake	Intel	5	47	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	N/A	
MCU_Motor _Error_Grade	Fault level	Intel	5	41	cycle	2	unsigned	1.0	0.0	0.0	3.0	0x00	0x3	N/A	
Motor_ ControlTemp	Controller temperature	Intel	4	32	cycle	8	unsigned	1.0	-50.0	-50.0	205.0	0x00	0xff	度	
Motor_Idc	BUS current	Intel	2	16	cycle	16	unsigned	0.1	-1000.0	-1000.0	5553.5	0x00	0xffff	A	
Motor_Udc	BUS voltage	Intel	0	0	cycle	16	unsigned	0.1	0.0	0.0	6553.5	0x00	0xffff	V	

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CAN ID List

CAN 1 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
VCU_MCU_Request	normal	0x160	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
MCU_Clamping_Brake_Req	Brake request VCU to MCU 0 : Release 1 : Hold brake	Intel	0	3	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
MCU_Speed_Req	[Speed command] - Requested rotation speed - Positive rotation speed means forward rotation - Negative rotation speed means reverse rotation	Intel	3	24	cycle	16	unsigned	1.0	-7000.0	-7000.0	7000.0	0x00	0xffff	RPM	
MCU_Torque_Req	[Requested torque] - Requested torque - Positive torque means forward rotation, - Negative torque means reverse rotation. - When the motor rotation direction is consistent with the positive and negative signs of the torque value, it is the motor power state, and vice versa, it is the power generation state.	Intel	1	8	cycle	16	unsigned	0.1	-1000.0	-1000.0	1000.0	0x00	0xffff	Nm	
MCU_DriveMode	Drive mode 0 : Torque 1 : Speed 2 : Torque ring 3 : Speed loop	Intel	0	1	cycle	2	unsigned	1.0	0.0	0.0	3.0	0x00	0x3		
MCU_VCU_Motor_Request_Valid	VCU Request Validity	Intel	0	0	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
MCU_Torque_Feedback	normal	0x10	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
MCU_ERRORCODE	Error code	Intel	7	56	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff	N/A	
MCU_MOTORTEMP	Motor temperature	Intel	6	48	cycle	8	unsigned	1.0	-50.0	-50.0	205.0	0x00	0xff	deg	
MCU_CURRENT	Motor current	Intel	4	36	cycle	12	unsigned	1.0	0.0	0.0	1000.0	0x00	0xfff	A	
MCU_TORQUE	Motor feedback torque	Intel	2	20	cycle	16	unsigned	0.1	-1000.0	-1000.0	5553.5	0x00	0xffff	Nm	
MCU_SPEED	Motor feedback speed	Intel	0	2	cycle	18	unsigned	1.0	-100000.0	-100000.0	162143.0	0x00	0x3ffff	rpm	
MCU_Shift	Stop	Intel	0	0	cycle	2	unsigned	1.0	0.0	0.0	3.0	0x00	0x3	N/A	

CAN ID List

CAN 1 ID

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
VCU_Meter_Req		normal			0x712			cycle			100			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
VCU_Meter_Req_Voltage	VCU voltage request to the meter	Intel	0	0	cycle	10	unsigned	1.0	0.0	0.0	1023.0	0x00	0x3ff	Intel	
VCU_Meter_Req_SOC	VCU requests the SOC power displayed by the meter	Intel	2	20	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff	Intel	
VCU_Meter_Req_Ready	0 : Not ready 1 : Ready	Intel	7	62	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	Intel	
VCU_Meter_Req_Mileage	VCU odometer request for the instrument	Intel	3	28	cycle	20	unsigned	1.0	0.0	0.0	1048575.0	0x00	0xfffff	Intel	
VCU_Meter_Req_ErrorCode	- VCU's fault code display request for the instrument - When 0 is sent, it means there is no fault, the maximum value is 999 - A fault code is sent every 5 seconds and displayed in polling.	Intel	6	52	cycle	10	unsigned	1.0	0.0	0.0	1023.0	0x00	0x3ff	Intel	
VCU_Meter_Req_Current	VCU requests the current value displayed by the meter	Intel	1	10	cycle	10	unsigned	1.0	-500.0	-500.0	500.0	0x00	0x3ff	Intel	
VCU_Meter_Req_Charge_State	0 : Not charged. 1 : Charged.	Intel	7	63	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1	Intel	

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
VCU_ICM_Req		normal			0x612			cycle			100			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
VCU_ICM_Req_msg5	- Content Accuracy - SOC 1% - Pack_Volt 1mV - Total current 1mA - Max unit voltage 0.001V - Highest monomer number - Min unit voltage 0.001V - Min monomer number - Max temperature 0.1℃ - Max temperature number - Min temperature 0.1℃ - Min temperature number - Fault code	Intel	4	32	cycle	32	signed	1.0	0.0	-2147483648.0	2147483647.0	0x00	0xffffffff		
VCU_ICM_Req_msg4	Fixed 0x01	Intel	3	24	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		
VCU_ICM_Req_msg3	- 0x61 SOC - 0x60 Pack_Volt - 0x60 Total Current - 0x61 Max single voltage - 0x61 Max Single Number - 0x61 Min Single Voltage - 0x61 Min Single Number - 0x61 Max temperature - 0x61 Max Temperature Number - 0x61 Min Temperature - 0x61 Min Temperature Number - 0x61 Fault code;	Intel	2	16	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		
VCU_ICM_Req_msg2	- 0x64 SOC - 0x40 Pack_Volt - 0x3E Total Current - 0x26 Max single voltage - 0x28 Max Single Number - 0x27 Min Single Voltage - 0x29 Min Single Number - 0x05 Max temperature - 0x40 Max Temperature Number - 0x06 Min temperature - 0x41 Min Temperature Number - 0x01 Fault code	Intel	1	8	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		
VCU_ICM_Req_msg1	Fixed 0x40	Intel	0	0	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		

CAN 1 ID

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Leng (Byte)
BMS_2B1h	normal	0x2b1	cycle	20	8

Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Sig Val Descrip
BMS_HVBatLowest TemCellNum	- High Voltage Battery Lowest Temperature Cell Number Valid value range 1 ~ 250 - 0xFE : Abnormal - 0xFF : Invalid	Intel	4	32	cycle	8	unsigned	1.0	0.0	0.0	250.0	0x00	0xff		
BMS_HVBatHighest TemCellNum	- High Voltage Battery highest Temperature Cell Number Valid value range 1 ~ 250 - 0xFE: Abnormal - 0xFF: Invalid	Intel	5	40	cycle	8	unsigned	1.0	0.0	0.0	250.0	0x00	0xff		
BMS_HVBat HighestTem	High Voltage Battery highest Temperature	Intel	1	8	cycle	8	unsigned	1.0	-40.0	-40.0	125.5	0x00	0xff	℃	
BMS_HVBat LowestTem	High Voltage Battery Lowest Temperature	Intel	0	0	cycle	8	unsigned	1.0	-40.0	-40.0	125.5	0x00	0xff	℃	

MSG Name	MSG Type	MSG ID	MSG Send Type	MSG Cycle Time (ms)	MSG Length (Byte)
BMS_A0h	normal	0xa0	cycle	20	8

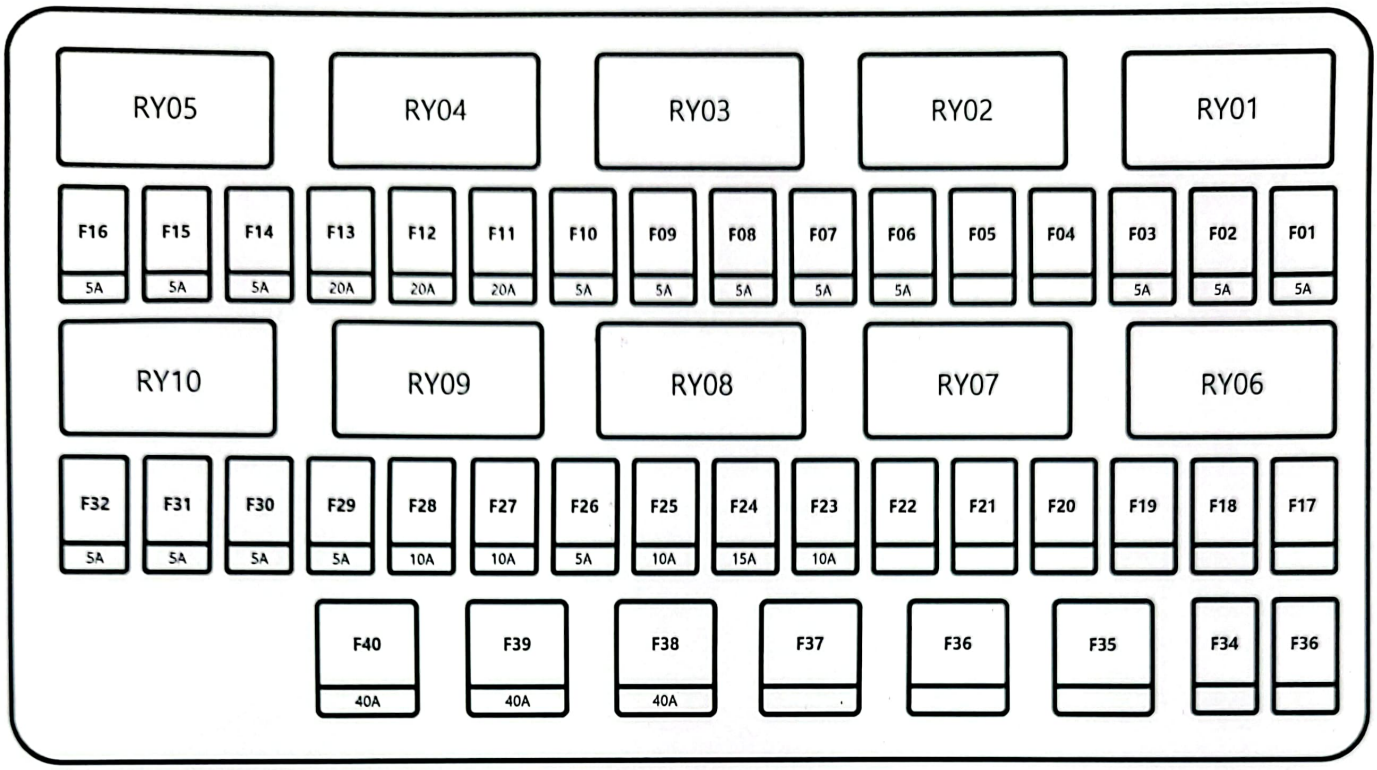
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Descripti
BMS_ HVDisplaySOH	Show SOH	Intel	7	56	cycle	8	signed	1.0	0.0	0.0	100.0	0x00	0xff	%	
BMS_Sys_Flt	error code	Intel	6	48	cycle	8	unsigned	1.0	0.0	0.0	255.0	0x00	0xff		
BMS_ Charge_StsCc	Slow charging connection status 0x 0x00 : Connect 0x01 : Disconnect	Intel	5	44	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
BMS_ Charge_StsCc2	Fast charging connection status 0x 0x00 : Connect 0x01 : Disconnect	Intel	5	43	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
BMS_Sys_Sts	0x0 : Normal 0x1 : Level 1 Fault(Alarm) 0x2 : Level 2 Fault(Power limit) 0x3 : Level 3 Fault (Power limit is 0, active shutdown)	Intel	5	40	cycle	3	unsigned	1.0	0.0	0.0	7.0	0x00	0x7		
BMS_HVBatSOC	High Voltage Battery State Of Charge High voltage battery SOC state	Intel	4	32	cycle	8	unsigned	0.4	0.0	0.0	100.0	0x00	0xff	%	
BMS_HVBatVol	High Voltage Battery Voltage High voltage battery current voltage	Intel	2	16	cycle	16	unsigned	0.1	0.0	0.0	1000.0	0x00	0xffff	V	
BMS_HVBatCrnt	High Voltage Battery Current High voltage battery current	Intel	0	0	cycle	16	unsigned	0.1	-1000.0	-1000.0	1000.0	0x00	0xffff	A	

CAN ID List

CAN 1 ID

MSG Name		MSG Type			MSG ID			MSG Send Type			MSG Cycle Time (ms)			MSG Length (Byte)	
BMS_1A2h		normal			0x1a2			cycle			20			8	
Signal Name	Signal Description	Byte Order	Start Byte	Start Bit	Signal Send Type	Bit Length	Date Type	Resolution	Offset	Signal Min. Value (phys)	Signal Max. Value (phys)	Signal Min. Value (Hex)	Signal Max. Value (Hex)	Unit	Signal Value Description
Charge_Relay_Sts	0x00 : Open 0x01 : Close charging relay status	Intel	6	51	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
AcDc_Relay_Sts	0x00 : Open 0x01 : Close Air conditioning relay status	Intel	6	52	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
BMS_WeakUpSig	BMS wake-up source 0x01 : Keyon 0x02 : A+ 0x04 : CP	Intel	0	0	cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
BMS_KeyOn	Normal voltage	Intel	2	16	cycle	8	unsigned	0.1	0.0	0.0	25.5	0x00	0xff	V	
BMS_Req _HvDown	BMS requests high voltage power off 0x00 : No request 0x01 : Requests high voltage power off	Intel	5	40	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Bms_SelfChk_Sts	BMS self-test status 0x00 : Self-test in progress 0x01 : Self-test successful 0x02 : Self-test failed	Intel	5	43	cycle	3	unsigned	1.0	0.0	0.0	7.0	0x00	0x7		
Heat_Relay_Sts	Heating relay status 0x00 : Open 0x01 : Close	Intel	5	46	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Precharge_Sts	Precharge status 0x00 : Invalid 0x01 : Precharge in progress 0x02 : Precharge successful 0x03 : Precharge failed	Intel	6	49	cycle	2	unsigned	1.0	0.0	0.0	3.0	0x00	0x3		
Charge_Sts	0x00 : Non-charging state 0x01 : Charging 0x02 : Heating 0x03 : Heating while charging 0x04 : Charging completed (When charging is completed, a charging failure is detected and charging is completed) 0x05 : Charging Fault	Intel	6	53	cycle	3	unsigned	1.0	0.0	0.0	7.0	0x00	0x7		
BMS_Heat_Beat	Heartbeat signal	Intel	7	56	cycle	4	unsigned	1.0	0.0	0.0	15.0	0x00	0xf		
Precharge _Relay_Sts	Precharge relay status 0x00 : Open 0x01 : Close	Intel	6	48	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		
Parent_Relay_Sts	Main positive relay status 0x00 : Open 0x01 : Close	Intel	5	47	cycle	1	unsigned	1.0	0.0	0.0	1.0	0x00	0x1		

Fuse Box 구성도



Number	*Name*	*Ampere*		*Number*	*Name*	*Ampere*
F01	MCU high voltage enable fuse	5A		RY1	Horn relay	
F02	MCU high voltage enable fuse	5A		RY2	Brake light relay	
F03	DC-DC High Voltage Enable Fuse	5A		RY3	Near light relay	
F06	BMS Power Fuse	5A		RY4	Right steering relay	
F07	Steering ignition fuse	5A		RY6	High voltage activation relay	
F08	Brake ignition signal fuse	5A		RY7	Reversing lamp relay	
F09	Remote emergency stop fuse	5A		RY8	Headlamp relay	
F10	IMU Fuses	5A		RY9	Left turn light relay	
F11	Front 12V Reserved Power Fuse	20A		RY10	ON RELAY	
F12	12V Reserved Power Fuse on board	20A				
F13	Rear 12V Reserved Power Fuse	20A				
F14	Low speed alarm fuse	5A				
F15	Remote start fuse	5A				
F16	Tire pressure sensor fuse	5A				
F23	VCU Power Fuse1	10A				
F24	Reset switch indicator power fuse	15A				
F25	VCU Power Fuse2	10A				
F26	Reversing lamp fuse	5A				
F27	High-light fuse	10A				
F28	Near-light fuse	10A				
F29	Brake light fuse	5A				
F30	Turn light fuse	5A				
F31	Horn power fuse	5A				
F32	ON gear ignition fuse	5A				
F38	EPS power supply fuse	40A				
F39	Brake power supply fuse	30A				
F40	ON-STAGE POWER SUPPLY FUSE	40A				

Unmanned Solution